

chondrosarcoma-mets-acetabular-fx-id h-unknown-y34r

Selected general biomarker report — biomarker screening coverage and gaps

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Selected general biomarker report

Which biomarkers on the selected panel this patient has been tested for, and which have never been measured. The panel covers targets established in this tumor type, targets relevant across a subset of tumor types, and biomarkers that predict benefit regardless of where the tumor started. An unmeasured biomarker below is a gap in the record, not a prediction that the patient carries it.

Surveyed: 14 biomarkers. **No usable result on file:** 14. **Measured but not decision-grade:** 0. **Already established:** 0.

Fourteen biomarkers were surveyed against this record, and not one has a usable result. Eleven were never measured. The other three, MAGE-A4, NY-ESO-1 and PRAME, cannot be assessed, because each reads against an HLA class I genotype that was never obtained. Two gaps are essential: MSI/dMMR and tumor mutational burden, the pair gating tumor-agnostic PD-1 blockade. Every row here is a hole in the record, not a statement about the tumor.

Closing all fourteen takes three orders. A comprehensive tumor DNA plus RNA panel, run on the dedifferentiated component, returns TMB, NTRK and RET fusions, BRAF V600E and the homologous recombination genes, and settles the IDH question the case has carried since intake. One block cut from the diagnostic biopsy covers the immunohistochemistry set: four mismatch-repair stains, PD-L1, B7-H3, HER2. A blood draw types HLA class I, and it should go first, since the allele decides whether the three cancer-testis antigen stains are worth the tissue; a non-A*02 type closes that class in one step. Those four rows sit with the target-validation track under their survey IDs. FAP is the outlier, imaging rather than tissue, worth ordering only if a FAP-directed trial is in play.

No gap needs a fresh biopsy: blood, the existing block (or the acetabular specimen if stabilisation surgery goes ahead), and one optional scan.

Expect negatives. MSI-H turned up in under 1 percent of sequenced sarcomas, TMB of 10 or more mutations per megabase in about 4 percent, and a TRK or RET fusion in a cartilaginous tumor would be a case report. The argument for testing is not that a positive is likely; it is the size of what a positive opens, approved tumor-agnostic drugs for a patient who otherwise has no approved targeted option, which fits her stated intent to pursue every possible one.

Gaps: biomarkers with no usable result on file

Microsatellite instability / mismatch-repair deficiency (MSI-H / dMMR)

Priority: essential **Scope:** Tumor-agnostic **Status:** Not measured

Primary site does not enter the question. A dMMR or MSI-H result gates approved PD-1 blockade in any solid tumor, which is why it belongs in a survey of a patient who has had no molecular testing at all.

Nothing is on file. MSI-H is genuinely rare in sarcoma, 0.7 percent in the largest clinically sequenced series and under 0.3 percent in an earlier 7,494-tumor cohort, concentrated in pleomorphic and radiation-associated subtypes, so a stable result is the expected one. The four stains are cheap, come back in days on the existing block, and a positive result would put approved checkpoint blockade in front of a patient who otherwise has no approved targeted option, along with a Lynch syndrome question for her and her relatives.

Recommended assay. Four-antibody mismatch-repair IHC (MLH1, PMS2, MSH2, MSH6), with MSI by PCR or NGS if IHC is equivocal (IHC · Diagnostic biopsy block if it holds enough dedifferentiated tumor; the acetabular specimen if stabilisation surgery goes ahead · 2-5 days)

Bundles with. One tissue block cut for the immunohistochemistry set

A positive result would open.

- pembrolizumab (PD-1 inhibitor · tumor-agnostic approval · NCT02628067)
- dostarlimab (PD-1 inhibitor · tumor-agnostic approval)

Next step. Order now

References. pmid:42531431; pmid:35705558; nct:NCT02628067

Tumor mutational burden, high (TMB-H, 10 or more mutations per megabase)

Priority: essential **Scope:** Tumor-agnostic **Status:** Not measured

Approved tumor-agnostically, so the cartilaginous origin is irrelevant to whether the result counts. It also comes free on the comprehensive panel this case needs for the IDH question.

No panel has been run, so TMB is unmeasured rather than low. Only 3.9 percent of 7,494 sarcomas reached 10 or more mutations per megabase in the largest profiling series, so a high value would be unusual here, but the number arrives at no extra cost on the same comprehensive panel that answers IDH, and crossing that threshold opens tumor-agnostic pembrolizumab. Specify a validated comprehensive panel on the requisition, because a small hotspot panel will still print a TMB figure that it is not sized to call and that figure should not drive a decision.

Recommended assay. TMB reported from a validated comprehensive genomic profiling panel, in mutations per megabase (NGS_panel · Diagnostic biopsy block if it holds enough dedifferentiated tumor; the acetabular specimen if stabilisation surgery goes ahead · 2-3 weeks)

Bundles with. Comprehensive tumor DNA plus RNA NGS panel run on the dedifferentiated component

A positive result would open.

- pembrolizumab (PD-1 inhibitor · tumor-agnostic approval · NCT02628067)

Next step. Bundle with planned testing

References. pmid:35705558; nct:NCT02628067

NTRK1 / NTRK2 / NTRK3 gene fusion

Priority: high **Scope:** Tumor-agnostic **Status:** Not measured

Approved without reference to primary site. Rare in chondrosarcoma, but the response rate when it is present is high enough that the test earns its place on any fusion panel being run for other reasons.

Not tested. A TRK fusion would be an uncommon finding in a cartilaginous tumor and the honest expectation is a

negative result, but three approved drugs sit behind it in any primary site and the assay costs nothing extra once a fusion panel is running. Ask for the RNA-based version: a DNA-only panel can miss a fusion whose breakpoint falls in a large intron, and the same RNA panel also answers RET and picks up HEY1-NCOA2 in passing.

Recommended assay. RNA-based fusion NGS panel covering NTRK1, NTRK2 and NTRK3 (RNA_seq · Diagnostic biopsy block if it holds enough dedifferentiated tumor; the acetabular specimen if stabilisation surgery goes ahead · 2-3 weeks)

Bundles with. Comprehensive tumor DNA plus RNA NGS panel run on the dedifferentiated component

A positive result would open.

- larotrectinib (TRK inhibitor · tumor-agnostic approval · NCT02576431)
- entrectinib (TRK / ROS1 / ALK inhibitor · tumor-agnostic approval)
- repotrectinib (next-generation TRK inhibitor · tumor-agnostic approval)

Next step. Bundle with planned testing

References. nct:NCT02576431

B7-H3 (CD276) surface protein expression

Priority: high **Scope:** Relevant across a tumor subset **Status:** Not measured

Pan-cancer surface antigen detectable in almost every bone and soft tissue sarcoma tested, with membranous H-scores rising with histological grade. Chondrosarcoma sits at the low end of the intensity range, so grade and subtype pull in opposite directions here and only a stain settles it.

No expression data exists. B7-H3 was detected in 96 percent of dedifferentiated chondrosarcomas in one series, and in 97 percent of 133 untreated bone and soft tissue sarcomas in another where H-scores rose with grade but chondrosarcoma sat at the low end of staining intensity, so presence is likely and intensity is the open question. Nothing behind this target is approved, so a positive stain buys trial candidacy rather than a prescription, and sponsors generally re-test centrally with their own antibody, which makes a local stain a triage tool rather than an eligibility document.

Recommended assay. B7-H3 (CD276) immunohistochemistry, membranous scoring (IHC · Diagnostic biopsy block if it holds enough dedifferentiated tumor; the acetabular specimen if stabilisation surgery goes ahead · 1-2 weeks)

Bundles with. One tissue block cut for the immunohistochemistry set

A positive result would open.

- ifinatamab deruxtecan (B7-H3 ADC · trial only)
- enoblituzumab (anti-B7-H3 monoclonal antibody · trial only)
- B7-H3 CAR-T (CAR-T · trial only)

Next step. Bundle with planned testing

References. pmid:33912442; pmid:42251292; pmid:39478506

HLA class I genotype (4-digit)

Priority: high **Scope:** Relevant across a tumor subset **Status:** Not measured

Host genotype, not tumor biology, so the primary site is beside the point. Without it the three cancer-testis antigen rows below cannot be assessed at all, and one of those antigens carries an approved TCR-T product in another sarcoma.

Never done, and it is the cheapest gap on this list to close: a blood draw with no tissue cost, which matters when the tumor block may be needed for several other tests. An HLA-A02:01 type *would make the cancer-testis antigen rows worth staining for*; a non-A02 type closes that whole class in one step and saves the tissue. Ordering it early also removes a several-week delay later, when a TCR-T referral would otherwise start from scratch.

Recommended assay. HLA class I genotyping at 4-digit (allele-level) resolution (hla_typing · Whole blood or buccal swab; no tumor tissue and no biopsy · 1-2 weeks)

Bundles with. Blood draw for HLA class I genotyping

A positive result would open.

- HLA-restricted TCR-T and ImmTAC agents (TCR-T / bispecific TCR · trial only)

Next step. Order now

References. pmid:27312065

PD-L1 protein expression

Priority: high **Scope:** Relevant across a tumor subset **Status:** Not measured

Chondrosarcoma is usually PD-L1 negative, but the dedifferentiated subtype is the exception: PD-L1 positivity clusters there, confined to the dedifferentiated component, alongside T-cell infiltrate and retained HLA class I. The stated histology is what makes this worth a stain instead of a reflex negative.

Untested. In the reference series PD-L1 was positive in 9 of 22 and then 11 of 21 dedifferentiated chondrosarcomas across two cohorts, and in none of 119 conventional, 19 mesenchymal or 20 clear cell tumors, with staining confined to the dedifferentiated component and tracking with tumor-infiltrating lymphocytes; a second group reported PD-L1 in a third of dedifferentiated tumors with CD8 T cells present in 90 percent of them. A positive result does not unlock an approved drug, since no sarcoma indication is keyed to PD-L1, so treat it as trial-enrichment information and make sure the section stained actually contains dedifferentiated tissue, or the result will be falsely negative.

Recommended assay. PD-L1 IHC with clone and scoring recorded explicitly, read on a section containing the dedifferentiated component (IHC · Diagnostic biopsy block if it holds enough dedifferentiated tumor; the acetabular specimen if stabilisation surgery goes ahead · 3-7 days)

Bundles with. One tissue block cut for the immunohistochemistry set

A positive result would open.

- PD-1 / PD-L1 inhibitors (checkpoint inhibitor · trial only · NCT02301039)

Next step. Bundle with planned testing

References. pmid:27312065; pmid:33912442; pmid:28988646; nct:NCT02301039

BRAF V600E mutation

Priority: medium **Scope:** Tumor-agnostic **Status:** Not measured

Actionable across primary sites, so the histology does not decide whether the result counts. Very rarely reported in cartilaginous tumors, and covered by any panel already being run for IDH.

Not tested, and unlikely to be positive in a cartilaginous tumor. It rides free on the panel already recommended for IDH, and if it is positive an approved tumor-agnostic regimen follows, so the only wrong move is ordering it as a standalone test. Non-V600E BRAF alterations do not qualify for this indication.

Recommended assay. BRAF codon 600 coverage on the comprehensive tumor NGS panel (NGS_panel · Diagnostic biopsy block if it holds enough dedifferentiated tumor; the acetabular specimen if stabilisation surgery goes ahead · 2-3 weeks)

Bundles with. Comprehensive tumor DNA plus RNA NGS panel run on the dedifferentiated component

A positive result would open.

- dabrafenib plus trametinib (BRAF plus MEK inhibitor · tumor-agnostic approval)

Next step. Bundle with planned testing

HER2 protein overexpression (IHC 3+)

Priority: medium **Scope:** Tumor-agnostic **Status:** Not measured

Trastuzumab deruxtecan is approved for IHC 3+ tumors in any primary site. HER2 overexpression is a carcinoma phenomenon and is not an expected finding in a mesenchymal tumor.

Not tested, and a 3+ result in a dedifferentiated chondrosarcoma would be distinctly unusual. One stain on a block already at the microtome is close to free, and the approved option behind a 3+ result is real, which is the whole argument for including it. If it comes back 1+ or 2+, that is HER2-low and does not satisfy the tumor-agnostic indication; 2+ reflexes to ISH before anything is concluded.

Recommended assay. HER2 IHC, only if a block is already being cut for the other stains (IHC · Diagnostic biopsy block if it holds enough dedifferentiated tumor; the acetabular specimen if stabilisation surgery goes ahead · 3-7 days)

Bundles with. One tissue block cut for the immunohistochemistry set

A positive result would open.

- trastuzumab deruxtecan (HER2-directed ADC · tumor-agnostic approval)

Next step. Bundle with planned testing

RET gene fusion

Priority: medium **Scope:** Tumor-agnostic **Status:** Not measured

Selpercatinib is approved for a RET fusion in any solid tumor. Essentially unreported in chondrosarcoma, but it shares an assay with the NTRK question.

Untested, and a RET fusion in a chondrosarcoma would be a case report. It is on the same RNA fusion panel as NTRK at no marginal cost, and the approved drug behind it is a strong one, so the sensible handling is to let it ride and not to order it separately. Note the distinction from a RET point mutation, which is a different, thyroid-specific indication.

Recommended assay. RET coverage on the RNA-based fusion NGS panel (RNA_seq · Diagnostic biopsy block if it holds enough dedifferentiated tumor; the acetabular specimen if stabilisation surgery goes ahead · 2-3 weeks)

Bundles with. Comprehensive tumor DNA plus RNA NGS panel run on the dedifferentiated component

A positive result would open.

- selpercatinib (selective RET inhibitor · tumor-agnostic approval · NCT03157128)

Next step. Bundle with planned testing

References. nct:NCT03157128

Fibroblast activation protein alpha (FAP)

Priority: medium **Scope:** Relevant across a tumor subset **Status:** Not measured

FAP is expressed on activated fibroblasts across solid tumors and on mesenchymal tumor cells themselves, and sarcomas are among the entities with the most consistent FAPI PET uptake. Uniquely on this list it can be assessed by imaging rather than by consuming tumor tissue.

Unmeasured. Everything behind FAP is early-phase, so this opens trial doors and nothing else, and FAPI PET is expensive and not available everywhere. It earns a row because it is the one target here that can be read without spending tissue, and because a FAPI PET would also characterise the pulmonary nodules and the acetabular lesion in the same acquisition.

Recommended assay. 68Ga-FAPI PET/CT, or FAP immunohistochemistry if a block is already being cut (imaging · None for the PET route; archival block for the IHC route · 1-3 weeks, depending on scanner and tracer access)

A positive result would open.

- ¹⁷⁷Lu-FAP-2286 and related FAP radioligand therapy (radioligand · trial only)
- FAP-directed bispecifics and CAR-T (bispecific / CAR-T · trial only)

Next step. Order if tissue available

References. pmid:41714122

Homologous recombination deficiency / BRCA1 / BRCA2 pathogenic variant

Priority: medium **Scope:** Relevant across a tumor subset **Status:** Not measured

PARP inhibitor approvals sit in ovarian, breast, pancreatic and prostate cancer, not sarcoma, so a finding here would be a trial or off-label matter. The HR genes are on the comprehensive panel regardless, and a germline finding in a woman in her forties carries family consequences of its own.

Nothing on file. Homologous recombination deficiency turned up in 2.5 percent of sarcomas in a 7,494-tumor series and no PARP inhibitor is approved for sarcoma, so the therapeutic payoff is a trial rather than a prescription. The genes come with the panel anyway, and a somatic BRCA1 or BRCA2 hit would trigger germline confirmation that matters for her relatives independently of what it does for her treatment.

Recommended assay. HR gene coverage on the comprehensive tumor panel, with germline testing if a BRCA1/2 or PALB2 variant appears (NGS_panel · Diagnostic biopsy block if it holds enough dedifferentiated tumor; the acetabular specimen if stabilisation surgery goes ahead · 2-3 weeks)

Bundles with. Comprehensive tumor DNA plus RNA NGS panel run on the dedifferentiated component

A positive result would open.

- PARP inhibitors (PARP inhibitor · approved, other tumor type)

Next step. Bundle with planned testing

References. pmid:35705558

MAGE-A4 (HLA-A*02:01-restricted)

Priority: medium **Scope:** Relevant across a tumor subset **Status:** Not assessable yet

Afamitresgene autoleucel is an approved TCR-T therapy for MAGE-A4-positive, HLA-A*02-positive synovial sarcoma, which makes MAGE-A4 the one cancer-testis antigen in this panel with an approved product in a sarcoma. Expression outside synovial sarcoma and osteosarcoma is patchier and has not been well characterised in chondrosarcoma.

Cannot be assessed at all until an HLA type exists, which is a blood draw. In the one series that stained bone and soft tissue sarcomas for cancer-testis antigens, MAGE-A4 was found in about 40 percent of osteosarcomas and a third of undifferentiated pleomorphic sarcomas, while the chondrosarcoma cases showed little cancer-testis antigen expression overall, so the expectation here is modest. Both the approved product and the trials also require fitness for lymphodepletion, which is another reason the assumed ECOG of 2 needs to become a measured one.

Recommended assay. HLA class I genotyping first; MAGE-A4 IHC or RT-PCR only if HLA-A*02:01 is present (hla_typing · Blood for typing; archival block for the reflex stain · 1-2 weeks for typing, then 1-2 weeks for the stain)

Bundles with. Blood draw for HLA class I genotyping

A positive result would open.

- afamitresgene autoleucel (TCR-T · approved, other tumor type)
- uzatresgene autoleucel (TCR-T · trial only)

Next step. Reflex if positive

References. pmid:40152485

NY-ESO-1 (CTAG1B, HLA-A*02:01-restricted)

Priority: low **Scope:** Relevant across a tumor subset **Status:** Not assessable yet

The classical sarcoma cancer-testis antigen, with TCR-T programs built around synovial sarcoma and myxoid liposarcoma. Cartilaginous tumors were not part of that development story.

Unassessable without an HLA type, and low yield if one arrives: NY-ESO-1 expression was low across every sarcoma subtype in the largest cancer-testis antigen survey of bone and soft tissue tumors, and the clinical programs are confined to synovial sarcoma and myxoid liposarcoma. Include it as a reflex only, on the same block as the other stains, and drop it entirely if the type is not HLA-A*02:01.

Recommended assay. NY-ESO-1 IHC or RT-PCR, reflexed from an HLA-A*02:01 result (IHC · Archival block, reflex only · 1-2 weeks after typing)

Bundles with. Blood draw for HLA class I genotyping

A positive result would open.

- letetresgene autoleucel (TCR-T · trial only)

Next step. Reflex if positive

References. pmid:40152485

PRAME (HLA-A*02:01-restricted)

Priority: low **Scope:** Relevant across a tumor subset **Status:** Not assessable yet

PRAME-directed ImmTAC and TCR-T programs are moving quickly, mostly in melanoma, and sarcoma subsets do express the antigen. Chondrosarcoma sits at the low end of the reported range.

Cannot be called without an HLA type. When 13 chondrosarcomas were stained in a multi-subtype sarcoma series, 1 was PRAME-positive, against 48 percent of undifferentiated pleomorphic sarcomas, so a negative result is the likely one. The programs behind it are trial-only, which puts this at the bottom of the list even though the reflex stain itself is cheap.

Recommended assay. PRAME IHC or RT-PCR, reflexed from an HLA-A*02:01 result (IHC · Archival block, reflex only · 1-2 weeks after typing)

Bundles with. Blood draw for HLA class I genotyping

A positive result would open.

- brenetafusp (PRAME-directed bispecific TCR (ImmTAC) · trial only)
- IMA203 (TCR-T · trial only)

Next step. Reflex if positive

References. pmid:40152485

Panel targets out of scope for this tumor

67 of 72 panel targets were screened and set aside as having no plausible connection to this patient's tumor type or stated features: ALPPL2 · ANTXR1 · AXL · CA9 · CD19 · CD200 · CD37 · CD46 · CD70 · CDCP1 · CDH17 · CDH3 · CDH6 · CEACAM5 · CLDN18 · CLEC12A · CT83 · DLK1 · DLL3 · EGFR · EGFR (variant III) · ENPP3 · EPCAM · EPHA2 · EPHB4 · ERBB2 · ERBB3 · F3 · FOLH1 · FOLR1 · FZD7 · GPC3 · GPRC5D · GUCY2C · HAVCR1 · IL13RA2 · IL3RA · ITGA3 · ITGB6 · L1CAM · LGR5 · LILRB4 · LY6E · LYPD3 · MAGEA1 · MAGEC2 · MERTK · MET · MMP14 · MSLN · MUC1 · NECTIN4 · PODXL · PSCA · PTK7 · ROR1 · SEZ6 · SLC34A2 · SLC44A4 · SLITRK6 · SSTR2 · SSX2 · STEAP1 · TACSTD2 · TNFRSF10B · TNFRSF17 · TSPAN8